

Program 6: 2D Matrix Multiplication

Aim

To write a C program to multiply two matrices.

Theory

Matrix A of order $r_1 \times c_1$ can be multiplied with matrix B of order $r_2 \times c_2$ only if $c_1 = r_2$ (columns of A equal rows of B).

The product C has order $r_1 \times c_2$.

Each element is the sum of products of row i of A with column j of B: $C[i][j] = A[i][0]*B[0][j] + A[i][1]*B[1][j] + \dots + A[i][c_1-1]*B[c_1-1][j]$.

Matrix multiplication is **not commutative** ($A \times B$ is generally not equal to $B \times A$) but is associative.

Time complexity: $O(r_1 \times c_1 \times c_2)$, which is $O(n^3)$ for square matrices.

Algorithm

1. Read the orders of A and B; if $c_1 \neq r_2$, print an error and stop.
2. Read the elements of A and B.
3. For $i = 0..r_1-1$ and $j = 0..c_2-1$: set $C[i][j] = 0$; for $k = 0..c_1-1$ add $A[i][k] * B[k][j]$ to $C[i][j]$.
4. Display matrix C.

C Source Code

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/*
 * Program : 2D Matrix Multiplication
 *
 * THEORY:
 * - A ( $r_1 \times c_1$ ) can be multiplied with B ( $r_2 \times c_2$ ) only if  $c_1 == r_2$ .
 * - The result C has order  $r_1 \times c_2$ .
 * -  $C[i][j] = \text{sum over } k \text{ of } (A[i][k] * B[k][j])$ 
 *   i.e. row i of A is multiplied with column j of B.
 * - Multiplication is NOT commutative ( $A*B \neq B*A$  in general).
 * - Time complexity:  $O(r_1 * c_1 * c_2) \rightarrow O(n^3)$  for square matrices.
 */
#include <stdio.h>
#define MAX 10

int main() {
    int A[MAX][MAX], B[MAX][MAX], C[MAX][MAX];
    int r1, c1, r2, c2;

    printf("Enter rows and columns of matrix A: ");
    scanf("%d %d", &r1, &c1);
    printf("Enter rows and columns of matrix B: ");
    scanf("%d %d", &r2, &c2);

    if (c1 != r2) {
        printf("Multiplication not possible (columns of A must equal rows of B).\n");
        return 1;
    }

    printf("Enter elements of matrix A:\n");
    for (int i = 0; i < r1; i++)
        for (int j = 0; j < c1; j++)
            scanf("%d", &A[i][j]);

    printf("Enter elements of matrix B:\n");
    for (int i = 0; i < r2; i++)
```

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        for (int j = 0; j < c2; j++)
            scanf("%d", &B[i][j]);

    for (int i = 0; i < r1; i++)
        for (int j = 0; j < c2; j++) {
            C[i][j] = 0;
            for (int k = 0; k < c1; k++)
                C[i][j] += A[i][k] * B[k][j];
        }

    printf("\nResult (A x B) of order %d x %d:\n", r1, c2);
    for (int i = 0; i < r1; i++) {
        for (int j = 0; j < c2; j++)
            printf("%5d", C[i][j]);
        printf("\n");
    }
    return 0;
}

```

Sample Input

Input: A is 2 x 3 = [1 2 3; 4 5 6], B is 3 x 2 = [7 8; 9 10; 11 12].

Sample Output (program run with the input above)

Enter rows and columns of matrix A: Enter rows and columns of matrix B: Enter elements of matrix A:
Enter elements of matrix B:

Result (A x B) of order 2 x 2:

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    58   64
   139  154

```